

Hafiz Muhammad Deen

hafizaxc@gmail.com | [in muhammaddeen](#) | [Portfolio site](#)

[muhammad](#) | +92-308-0026860 | Mianwali, Pak

Skills

Languages: C++, Java, Python, JavaScript, Assembly Language, SQL.

Technologies & Tools: React, HTML, CSS, Node.js, Flutter, Dart, React Native, FastAPI, OpenCV, Gemini API, VS Code, Firebase, Cisco Packet Tracer, Git, GitHub.

Core Concepts: Digital Image Processing, Object-Oriented Programming (OOP), Data Structures, Computer Networks, System Architecture.

Education

University of Mianwali | Mianwali, Pakistan

Oct 2023 - Present

CGPA: 3.34/4

B.S. in Computer Science

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Web Technologies, Computer Networks, Artificial Intelligence, Digital Image Processing, Database Systems, Statistics and Probability.

Project Work

• InnoVarsity (AI-Driven Science Fair Platform) | Final Year Project (Ongoing):

- Solved the problem of subjective human bias and manual grading in academic exhibitions.
- Upload project assets (Poster, PDF, Video) to get an automated, objective AI-generated score.
- Uses Microservices architecture, OpenCV, FAISS, and Sentence-transformers.
- Frontend build in React Native and React JS.
- Backend in Node.js (API Gateway) and FastAPI Python (AI Engine).

• InnoVarsity-Poster-AI-Evaluator (Standalone AI Module): ([View](#))

- Solved the problem of subjective and inconsistent manual grading of academic research posters.
- Evaluates uploaded posters autonomously to generate an objective score based on Aesthetics, Structure, and Academic Depth.
- Uses OpenCV for blur detection (Laplacian), contrast analysis, and dynamic structural block detection via Morphological transformations.
- Integrates Tesseract OCR to extract text, calculate lexical density, and verify domain-specific technical vocabulary.
- Developed entirely in Python using OpenCV, PyTesseract, and NumPy.

• Heap Sort Visualizer (Interactive Algorithm Simulator) | Full Stack Project: ([View](#))

- Solved the complexity of learning algorithms through real-time, interactive visualization.
- Features dynamic array generation and adjustable speeds to track Max-Heap sorting mechanics.
- Uses asynchronous programming (Async/Await) for smooth, non-blocking animations.
- Frontend built in React JS (Vite).
- Backend in Node.js for server-side execution.

- **Dynamic Programming Visualizer (Algorithm Optimizer) | Full Stack Project: ([View](#))**

- Solved the complexity of learning advanced DP concepts (Memoization & Tabulation) via interactive visualization.
- Features real-time state tracking to visually demonstrate overlapping subproblems and optimal substructure.
- Uses asynchronous programming (Async/Await) for step-by-step execution rendering.
- Frontend built in React JS (Vite).
- Backend in Node.js for server-side execution.

- **VIC Modern Hotel - Enterprise Network: ([View](#))**

- Solved the problem of secure and isolated communication across a multi-floor organization.
- Simulates a real-world enterprise network with departmental isolation and wireless access.
- Uses VLAN segmentation, Inter-VLAN routing, OSPF, and DHCP.
- Security handled via SSH remote access and switch port security.
- Designed and tested entirely in Cisco Packet Tracer.

- **Certifications & Courses:**

- **Artificial Intelligence Foundation: ([View](#))** Completed comprehensive coursework focusing on AI concepts and problem-solving techniques.
- **Advanced C++ Programming: ([View](#))** Strengthened logic building, memory management, and algorithmic thinking.
- **Affiliate Marketing: ([View](#))** Gained insights into digital marketing, user acquisition, and online business models.

Achievements & Societies

- **NexGen Coding Society - Active Member:**

- Actively participating in a community of tech enthusiasts to discuss modern software development trends.
- Collaborating with peers to enhance problem-solving skills and participate in campus-level tech initiatives.